

World Cup Predictor Dataset Build

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.6.1

Warning: package 'forcats' was built under R version 4.6.1

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(worldcup)
```

```
# Predictor 1: Historical World Cup Win Rate Difference
```

```
team_match_history <- bind_rows(

  matches |>
    transmute(
      tournament_id,
      match_id,
      match_date,
      team_code = home_team_code,
      team_name = home_team_name,
```

```

      opponent_code = away_team_code,
      goals_for = home_team_score,
      goals_against = away_team_score,
      win = as.integer(home_team_score > away_team_score)
    ),

matches |>
  transmute(
    tournament_id,
    match_id,
    match_date,
    team_code = away_team_code,
    team_name = away_team_name,
    opponent_code = home_team_code,
    goals_for = away_team_score,
    goals_against = home_team_score,
    win = as.integer(away_team_score > home_team_score)
  )
)

team_match_history <- team_match_history |>
  mutate(
    year = as.integer(
      str_remove(tournament_id, "WC-")
    )
  )
)

teams_by_tournament <- team_match_history |>
  distinct(
    year,
    team_code,
    team_name
  )
)

wc_win_rate_by_tournament <- map_dfr(
  sort(unique(teams_by_tournament$year)),
  function(target_year) {

    current_teams <- teams_by_tournament |>
      filter(year == target_year)

    prior_history <- team_match_history |>

```

```

    filter(year < target_year)

current_teams |>
  left_join(
    prior_history |>
      group_by(team_code) |>
      summarise(
        prior_wc_matches = n(),
        prior_wc_wins = sum(win),
        .groups = "drop"
      ),
    by = "team_code"
  ) |>
  mutate(
    prior_wc_matches = replace_na(prior_wc_matches, 0L),
    prior_wc_wins = replace_na(prior_wc_wins, 0),
    wc_win_rate = if_else(
      prior_wc_matches > 0,
      prior_wc_wins / prior_wc_matches,
      0
    )
  ) |>
  select(
    year,
    team_code,
    team_name,
    prior_wc_matches,
    prior_wc_wins,
    wc_win_rate
  )
}
)

```